

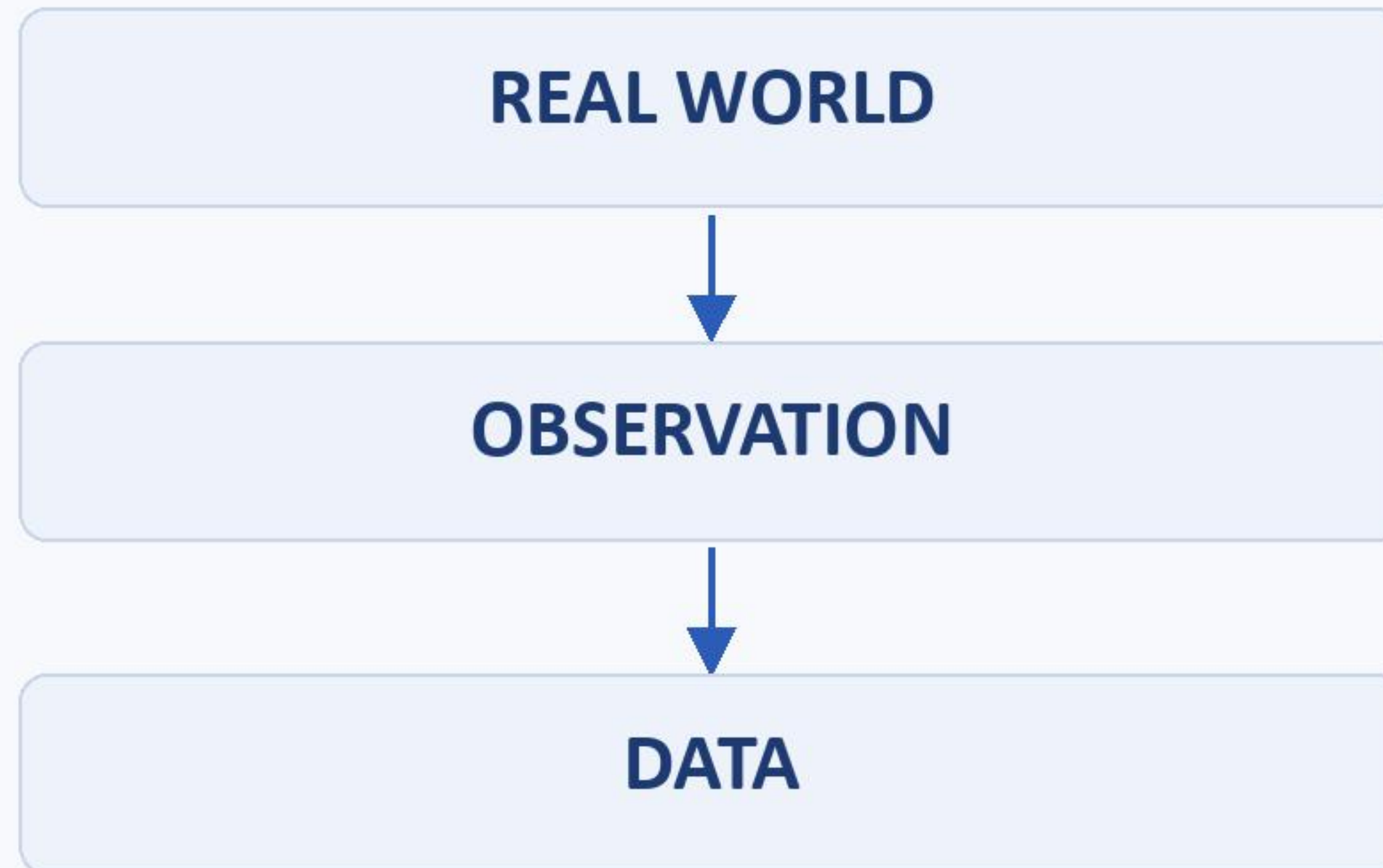
How Computers Represent Information

From Information → Bits → Binary

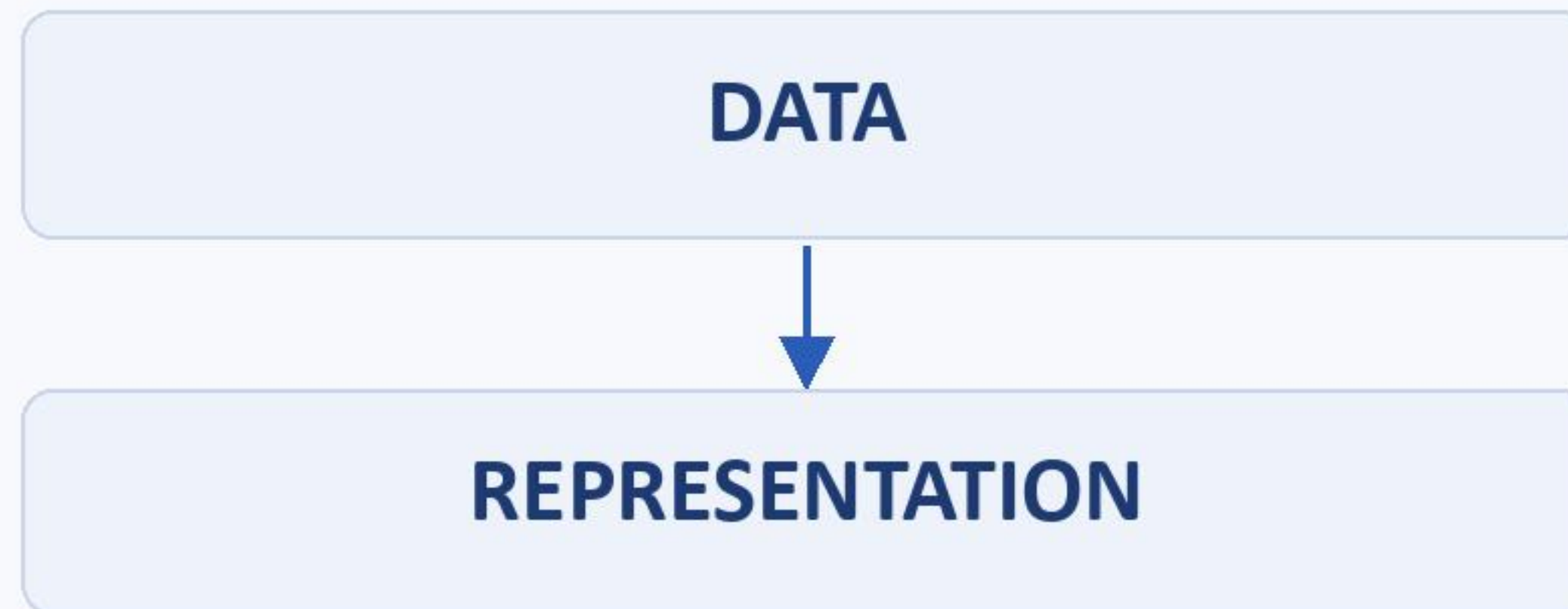
THE BIG QUESTION

If computers work with information like text, images, and sound, how do they actually represent that information?

What Happens Next?



Data Needs a Representation



Before a computer can store or process information, it needs a form of that information it can work with.

Many Kinds of Information

NUMBER

TEXT

IMAGE

SOUND

VIDEO

**How can the same kind of computer
work with all of these?**

A Simple Idea: Two States

LIGHT

OFF | ON

DOOR

CLOSED | OPEN

QUESTION

NO | YES

Digital

Information represented using distinct, clearly distinguishable states.

CONTINUOUS RANGE



could be anywhere along here

VS.

TWO CLEAR STATES



always clearly one or the other

THE KEY BUILDING BLOCK

Bit

BIT

BINARY DIGIT

ONE OF TWO POSSIBLE STATES

One Bit

ONE BIT

0 | 1

two possible states

THINK ABOUT IT

**If one switch has two states,
how many different messages
can it represent?**

A real yes/no answer only needs two options, too.

Two Bits

00

01

10

11

Four possible combinations — not two more, but double the one-bit case.

Binary Combinations

1 BIT → 2

2 BITS → 4

3 BITS → 8

Each extra bit roughly doubles the number of possible combinations.

Binary Is a Representation System

**A BIT PATTERN IS NOT
"THE THING ITSELF."
IT REPRESENTS INFORMATION.**

Representation, Not the Real Thing

CLASS 03

“80% Full”

Represents a crowded canteen — it is not the canteen itself.

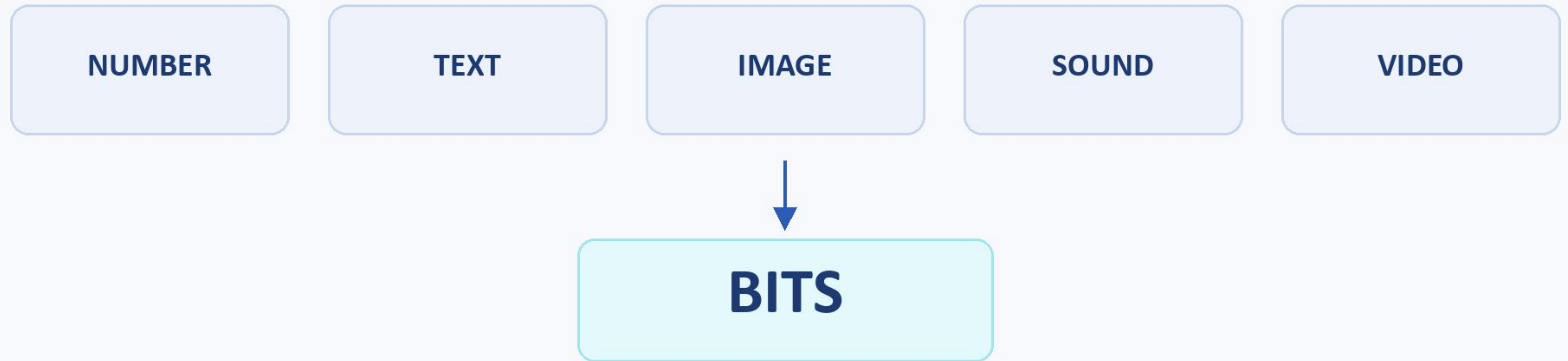
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CLASS 04

A Bit Pattern

Represents information for a computer — it is not the information itself.

Same Building Block, Every Time



Very different information — the same underlying building block.

STILL A LITTLE MYSTERIOUS

**How could millions of tiny
binary decisions possibly
describe an image?**

Same Marks, Different Meaning

CONTEXT A

“S O S”

→ a distress signal

=

CONTEXT B

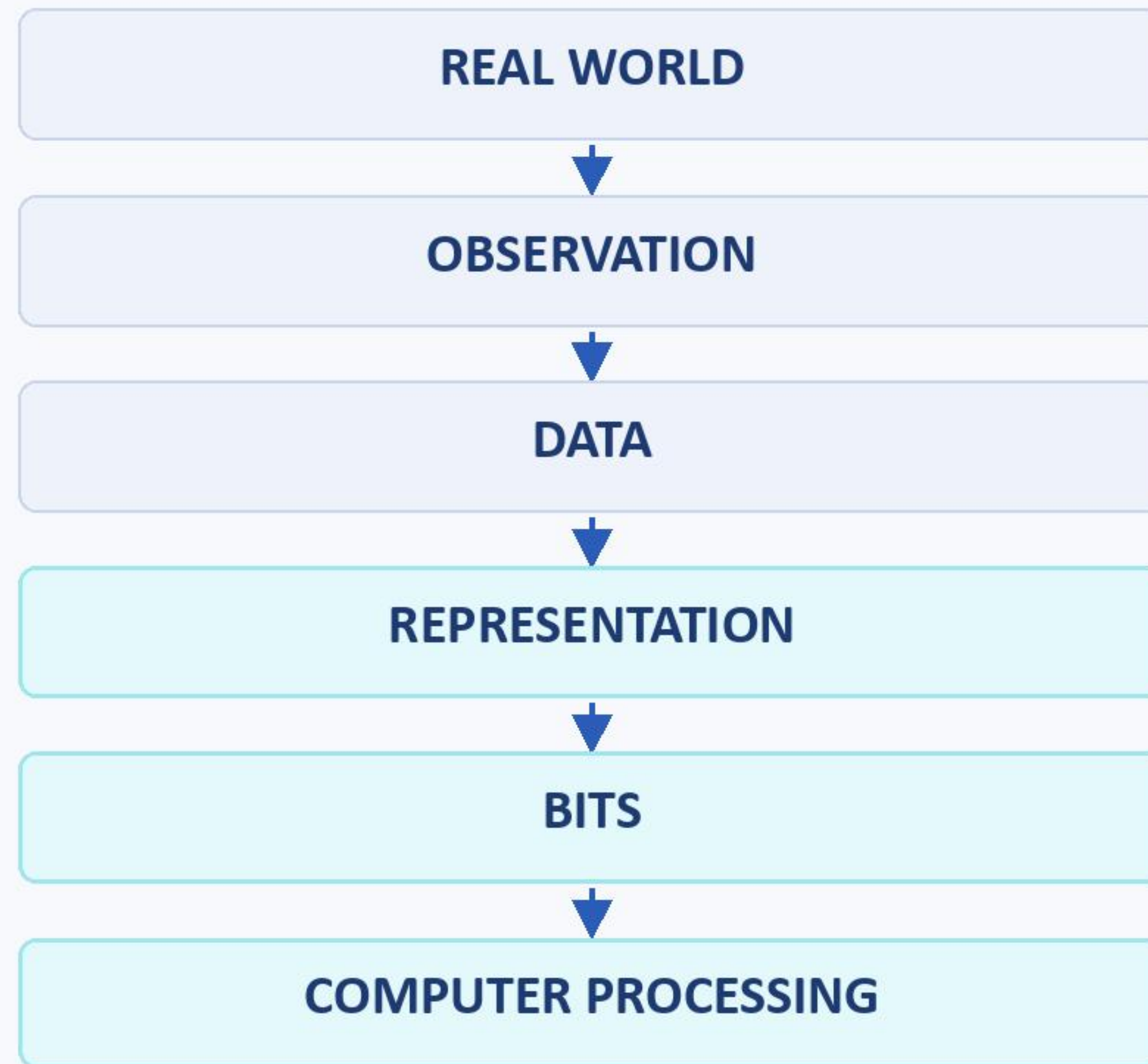
“S O S”

→ just someone's initials

Where Meaning Comes From

**MEANING COMES FROM
INTERPRETATION —
NOT FROM THE BITS THEMSELVES.**

THE HERO DIAGRAM



Cyan = new today · Card = already known from Class 03

Design a Tiny Binary Communication System

You have 2 bits — four combinations:

00 01 10 11

Invent a meaning for each one.

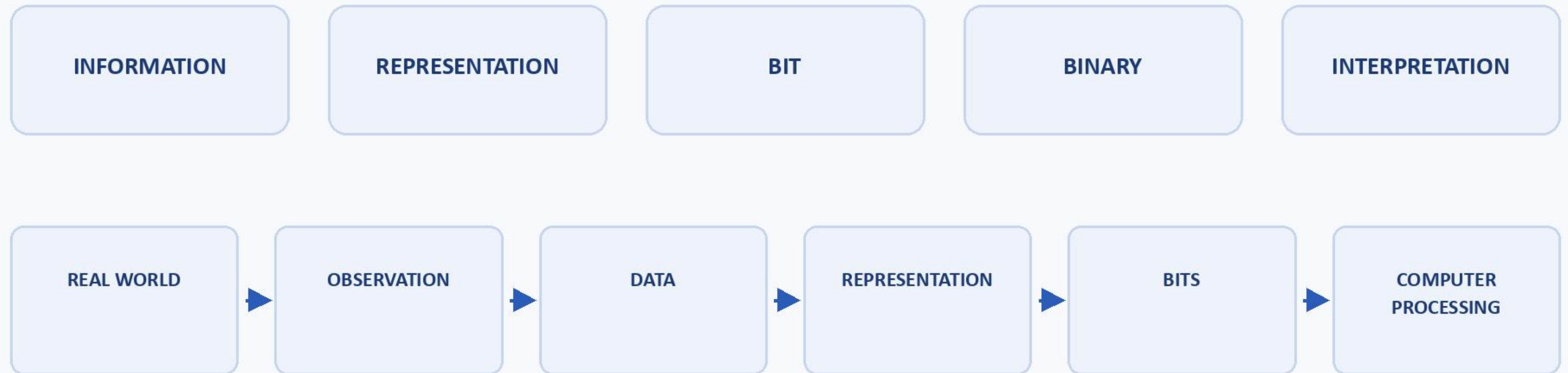
Example: A Tiny Color Code

Bits	Meaning
00	Red
01	Green
10	Blue
11	Yellow

THE KEY QUESTION

**Did the bits themselves
contain the meaning?**

Quick Recap



A QUESTION FOR NEXT TIME

If the same building blocks can represent numbers, text, images, and sound — what determines what those bits actually mean?

FINAL TAKEAWAY

“Computers do not need a different kind of machine for every kind of information. They can build many kinds of digital information from the same basic building block: bits.”